

Escape Into The Fascinating World Of CONTROL SIMULATION AND DESIGN

SHANOSH KUMAR

MPLAB X IDE

MPLAB X integrated development environment (IDE) is a must-have open source software program. It is based on NetBeans IDE from Oracle. This software consists of the most recent advancements, features and extensible architecture an engineer working on microchip, microcontrollers and digital signal controllers would expect.

Specially bundled into this DVD, the software runs on any platform such as Windows, Mac OS or Linux. Features are clubbed into the IDE, because it provides a single integrated environment to develop code for embedded microcontrollers.

ASEM-51

ASEM-51 is a two-pass macro assembler for Intel MCS-51 family of microcontrollers. ASEM-51 assembly language is based on the standard Intel syntax and implements conditional assemblies and macros, and includes file processing. The assembler can output object code in Intel-HEX or Intel OMF-51 format as well as a detailed list file. ASEM-51 package includes more than 200 8051 derivatives and a bootstrap program for MCS-51 target boards. This software also runs seamlessly on MS-DOS, Windows and Linux platforms.

Express PCB

The software facilitates designing and efficient manufacturing of double-sided and four-layered PCBs, thereby solving the manufacturing problem of achieving greater turnaround time for prototyping. The user interface and functionality is designed to be very easy, and requires only a couple of hours to get used to.

This software easily adapts itself to any platform but works seamlessly

Some popular resources

OpenCV. Also called open source computer vision, it facilitates computational efficiency for real-time applications such as camera calibration, image data manipulation, motion analysis and so on. The software is written in optimised C/C++ so that the library can take advantage of multi-core processing. It has C++, C, Python and Java interfaces. It runs seamlessly on Windows, Linux, Mac OS, iOS and Android platforms.

matplotlib. This is a Python library for making publication-quality plots using a syntax familiar to MATLAB users. This software helps you to generate plots, histograms, power spectra, bar charts, error charts, scatter plots and the like, using just a few lines of code. Outputs generated can also be exported into PDF, Postscript, SVG and PNG formats, as well as used for screen display. The user interface is designed to provide an object-oriented interface with functions familiar to MATLAB such as full control of line styles, font properties, axes properties, etc.

GNS3. A graphical network simulator, GNS3 allows you to design complex network topologies as well as configure virtual networks. You can run simulations or configure devices ranging from simple workstations to powerful Cisco routers.

on Windows, including the latest Microsoft Windows 10 edition. As a beginner, you could get started with drawing a schematic by inserting component footprints and dragging these into position. Connecting the pins by drawing traces is quite easy. As a professional, you can go about extensively designing complex boards that could be made production-ready with great ease.

Mango Automation

Mango Automation is supervisory control and data acquisition (SCADA) software for industrial control. Developers of this software have coded down its core in Java, and it can run on any platform that supports Java. The core provides all base services that this SCADA application uses, while modules could be plugged to the core instance by copying into a sub-directory to create a SCADA application.

Modules provide functionality such as connectivity to equipment using data sources, graphical user interfaces, dashboards, language translations, themes and so on. The core and modules of this software are supported by a Web based user interface on the foreground, and the application could be run from anywhere in the world through the Internet.

Sprint-Layout 6.0

Sprint-Layout is a perfect tool for creating single-sided and double-sided PCBs. The software comes with all functions that are necessary for board design. It supports professional features like export of Gerber files for professional manufacturing or for isolation milling with CNC machines.

Fedora Workstation

Fedora is an operating system based on Linux kernel. This operating system comes pre-installed with a wide range of software such as LibreOffice, Google Chrome, Java Iced Tea, Adobe Flash Player, Skype and Firefox. If additional software is required, there are software repositories that can be installed using DNF package manager or GNOME software.

You will find it similar to DebianBlends while working on Fedora, as it also facilitates custom variations of Fedora called Fedora Spins or Fedora Editions. These are built with specific sets of software packages that offer alternative desktop environments as well as supplement specific interests such as gaming, security, design, education, robotics and scientific computing. **EFY**

Shanosh Kumar is technology journalist at EFY. He is BCA from Bangalore University and MBA from Christ University, Bengaluru